Humanized TNFR2 Mouse

Strain Name: C57BL/6-*Tnfrsf1b* $^{em2(hTNFRSF1B)Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-190010

Tumor necrosis factor receptor 2 (TNFR2), also known as tumor necrosis factor receptor superfamily member 1B (TNFRSF1B), and CD120b, is one of major receptors of the cytokines, TNF and lymphotoxin-α. TNFR2 has proinflammatory effects too, but it also elicits strong anti-inflammatory activities and has protective effects on oligodendrocytes, cardiomyocytes, and keratinocytes.

Construction strategy

On the C57BL/6 background, the full-length coding sequence of human TNFR2 gene was placed immediately downstream of the start codon of the mouse endogenous Tnfr2 gene, followed by a poly(A) element gene via CRISPR/Cas9 mediated recombination. This guarantees an exclusive expression of human TNFR2 in the hTNFR2 mice.

Validation data

- **Validation of humanized TNFRSF1B(TNFR2) expression in the immune cells from Spleen**

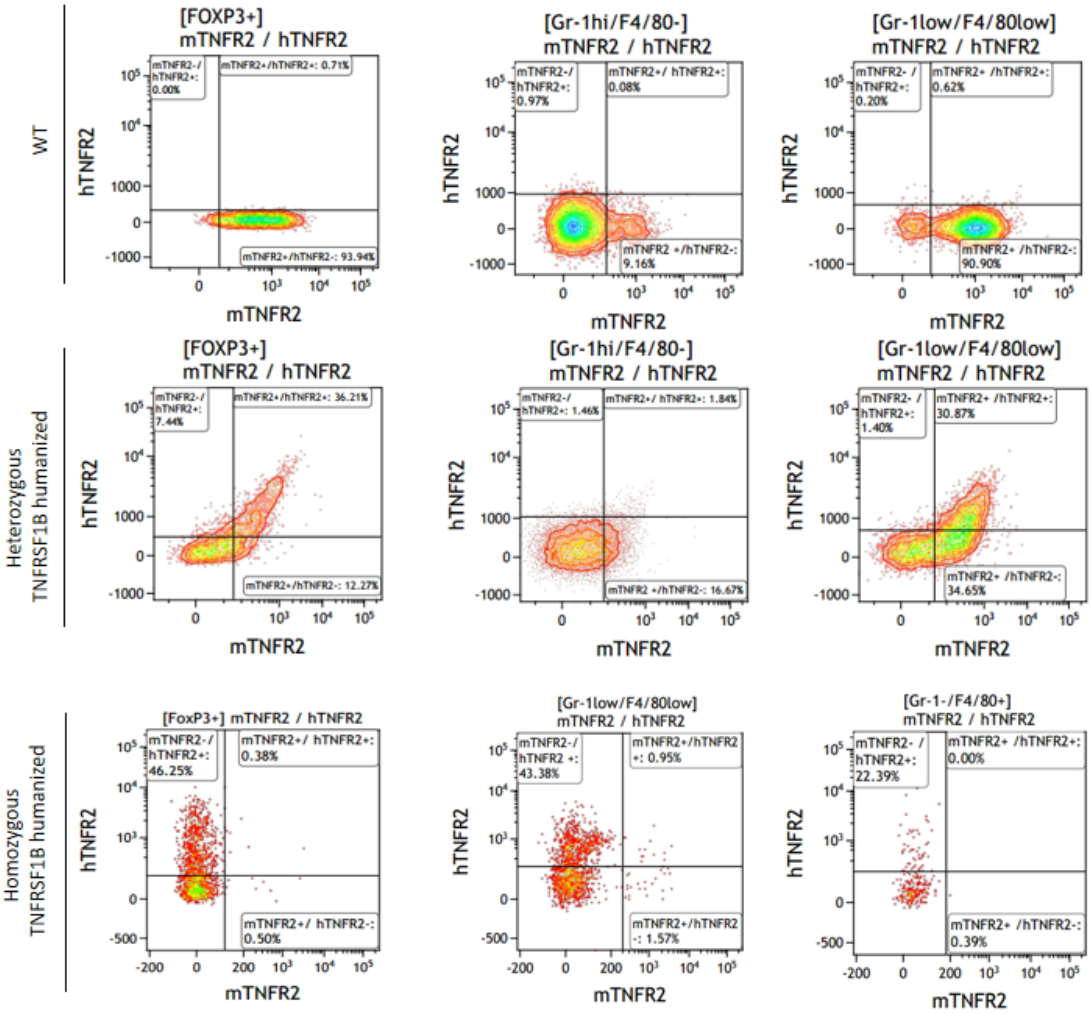

Figure 1. The expression of human TNFRSF1B in humanized TNFRSF1B mice was measured by FACS. In humanized TNFRSF1B mice, active expression of human TNFRSF1B was detected in Treg cells and Neutrophils isolated from spleen. In collaboration with CrownBio).

- **Validation of humanized TNFRSF1B(TNFR2) expression in the immune cells from Peripheral Blood**

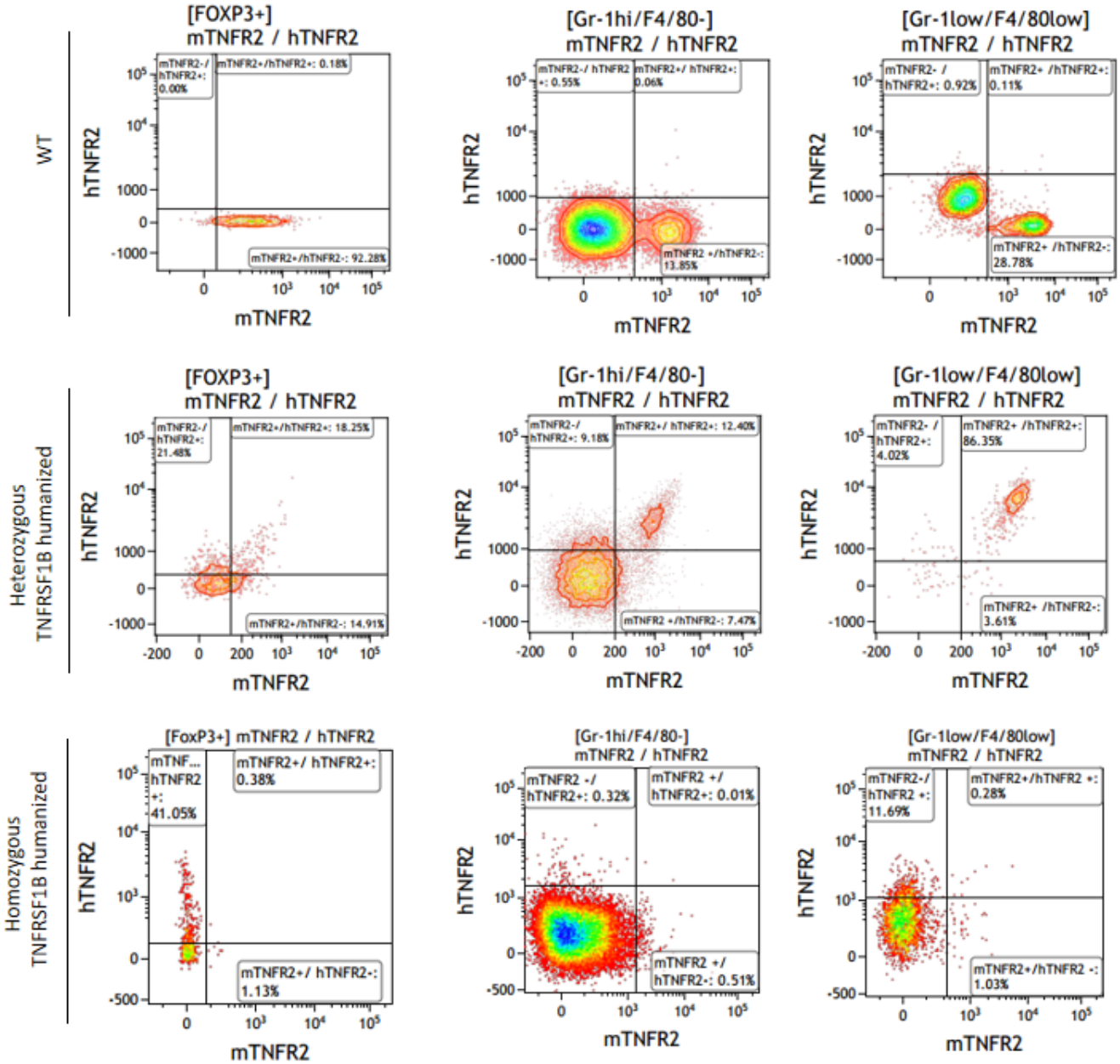

Figure 2. The expression of human TNFRSF1B in humanized TNFRSF1B mice was measured by FACS. In humanized TNFRSF1B mice, active expression of human TNFRSF1B was detected in Treg cells and Neutrophils isolated from peripheral blood cells. (In collaboration with CrownBio)

- **Validation of humanized TNFRSF1B(TNFR2) expression in the immune cells from Lymph node**

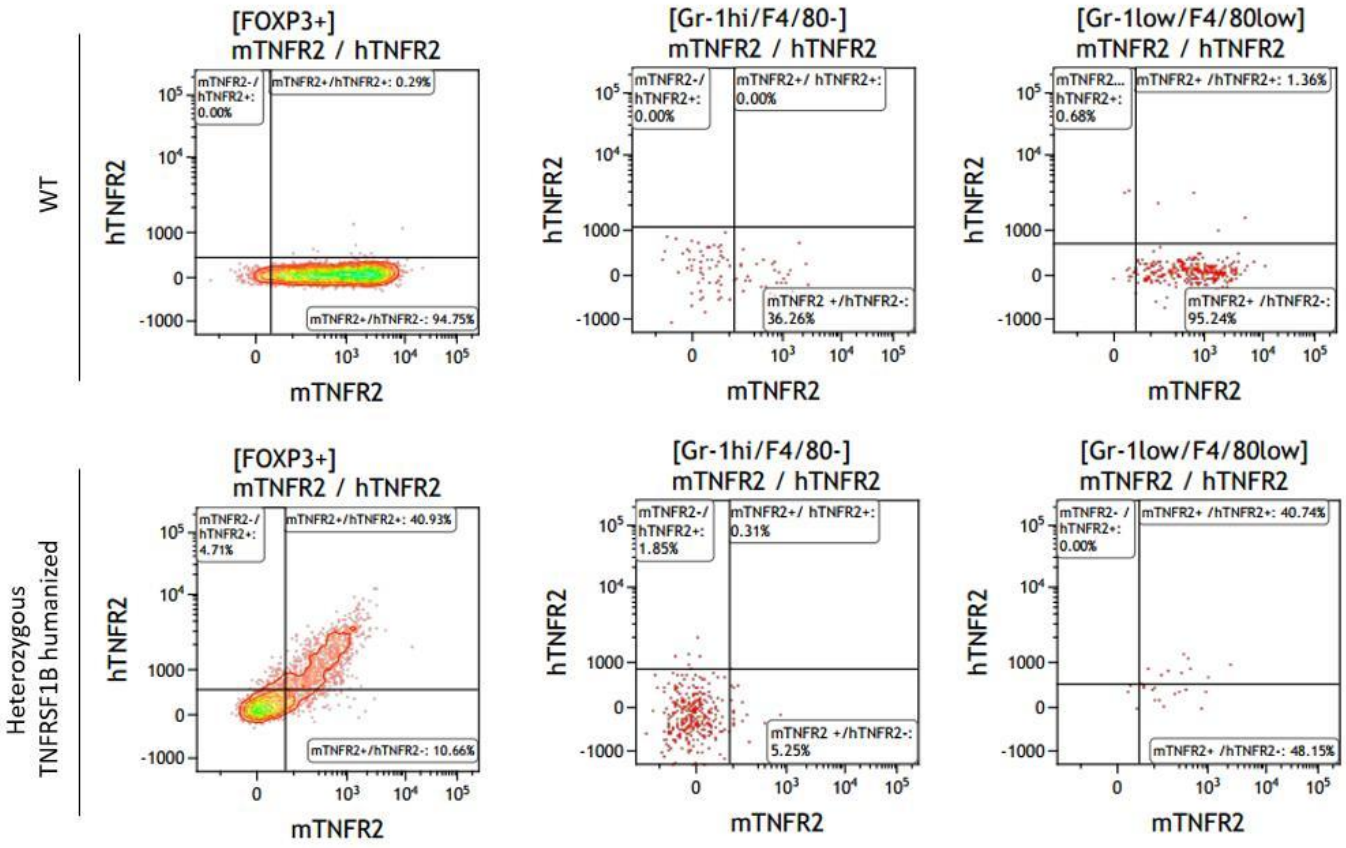

Figure 3. The expression of human TNFRSF1B in humanized TNFRSF1B mice was measured by FACS. In heterozygous humanized TNFRSF1B mice, active expression of human TNFRSF1B was detected in Treg cells isolated from Lymph node. (In collaboration with CrownBio)

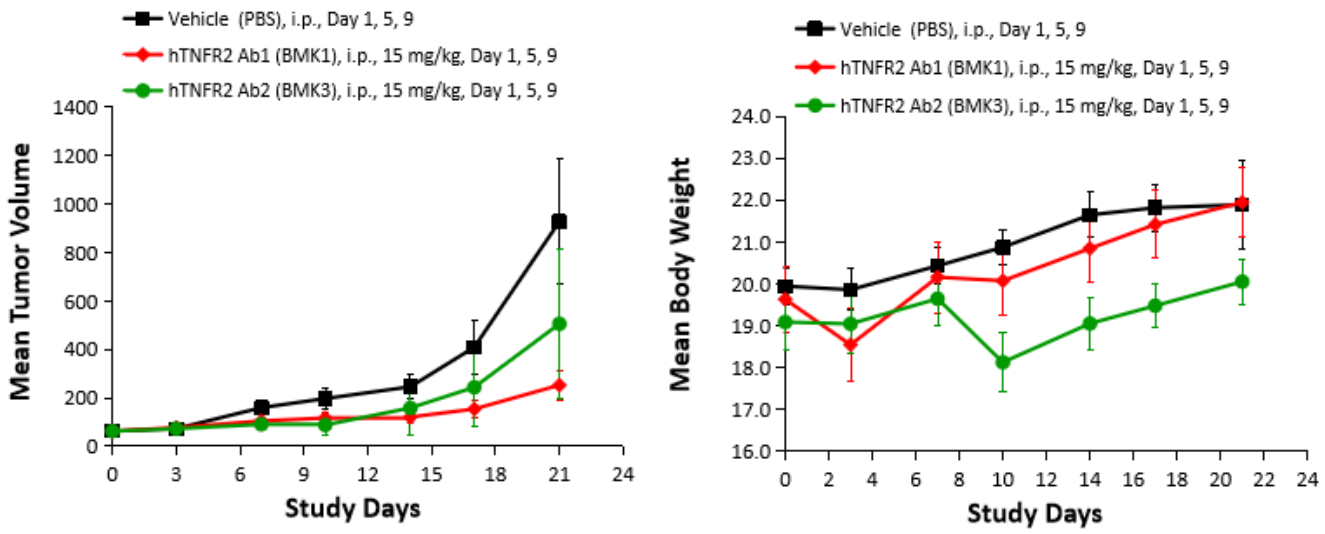

Figure 4. In vivo anti-tumor effect of anti-TNFRSF1B antibody, in a humanized mouse model of TNFRSF1B In vivo validation of anti-tumor efficacy in a MC38-OVA tumor-bearing model of humanized TNFRSF1B mice. Homozygous humanized TNFRSF1B mice were inoculated with MC38 colon cancer cells. The drug targeting human TNFRSF1B, showed a very significant anti-tumor effect, demonstrating that the humanized TNFRSF1B mice are a good in vivo model for validating the efficacy of antibodies targeting human TNFRSF1B. (In collaboration with CrownBio)

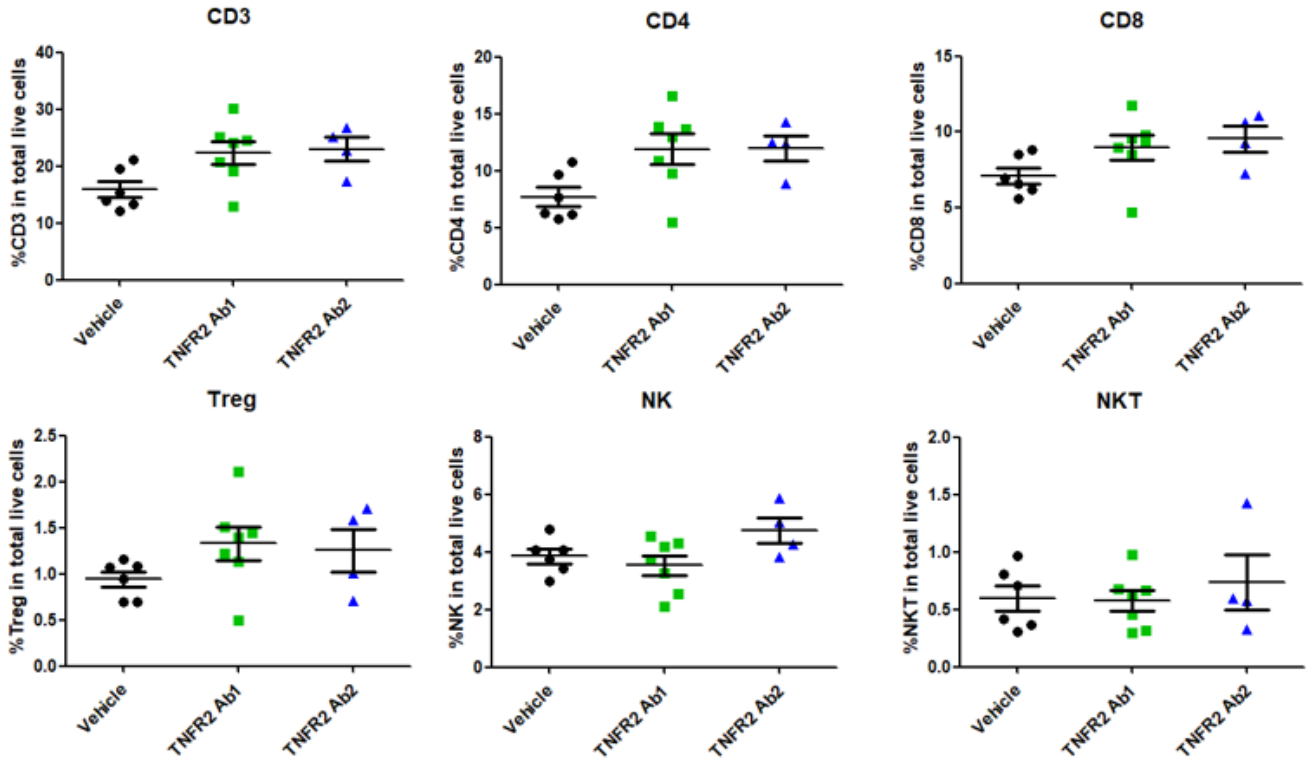

Figure 5. FACS for blood upon TNFRSF1B antagonist treatment. Addition of antagonistic hTNFRSF1B antibody can impair hTNF-induced Treg cells proliferation, while the agonistic hTNFRSF1B antibody alone can induce a modest proliferative response. (In collaboration with CrownBio)

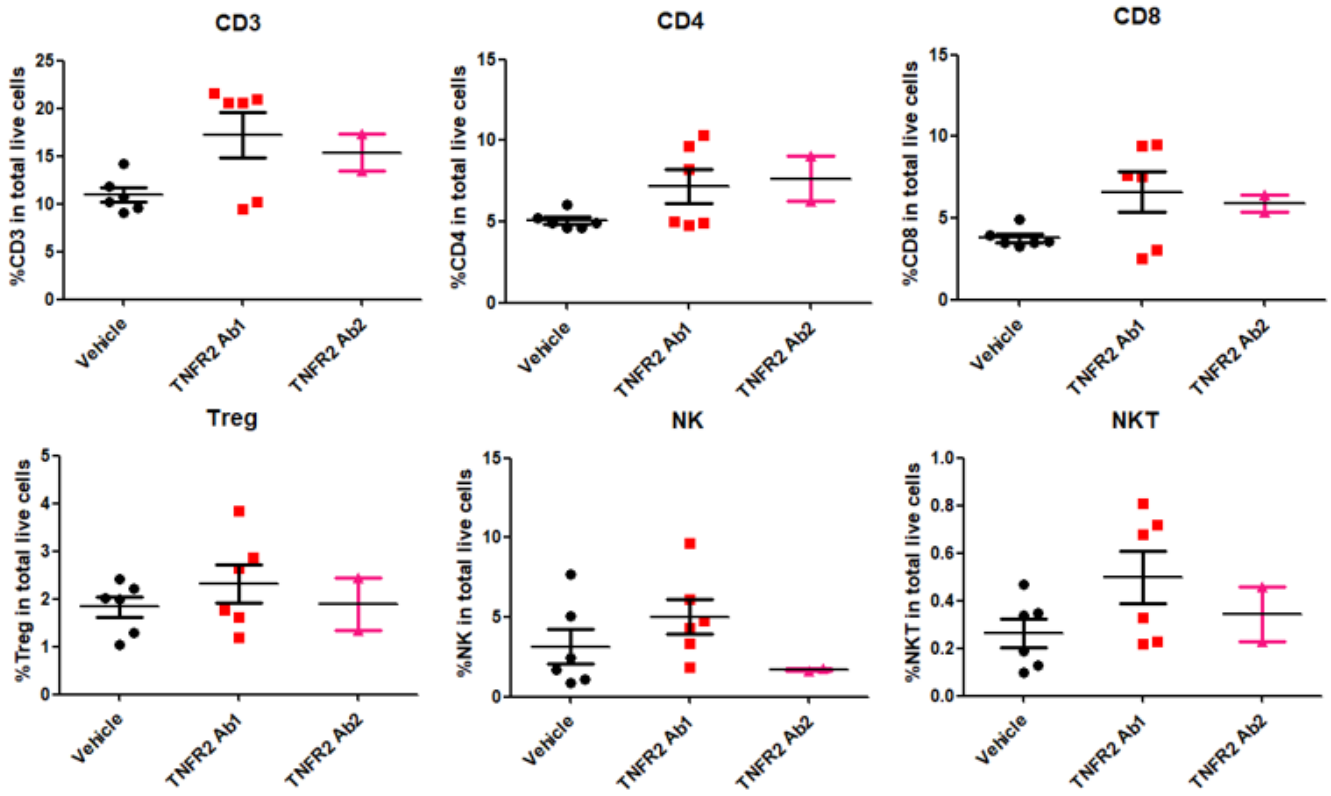

Figure 6. TIL analysis upon TNFR2 antagonist treatment.

1 tumor in hTNFR2 Ab1 treatment group decreased (1/7, TV=0)

2 tumors in hTNFR2 Ab2 treatment group decreased (2/7, TV=0; 3/7 died during dosing period)

(In collaboration with CrownBio)

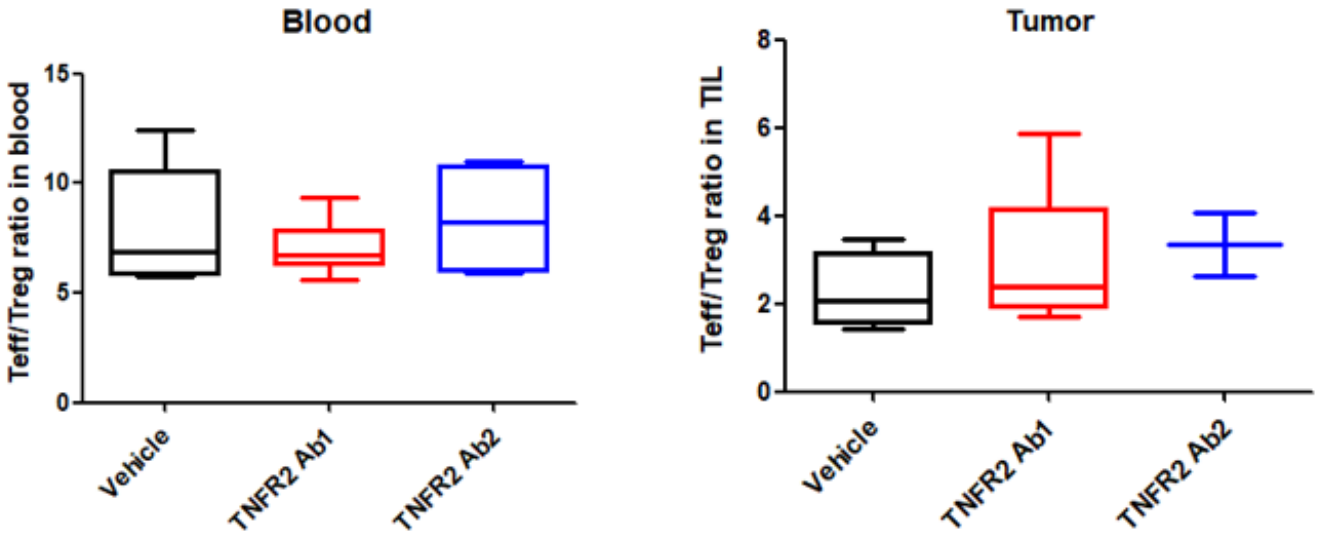

Figure 7. TNFRSF1B blockade increases the Teff/Treg ratios in tumor microenvironment. (In collaboration with CrownBio)

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com**
or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

ingenious targeting laboratory

2200 Smithtown Ave, Ronkonkoma, NY 11779

www.genetargeting.com • inquiry@genetargeting.com